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Q1 A1

4 points

Q1.1 (a)

1 point

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(a) Evaluate $\lim_{x \rightarrow -\infty} \sqrt{x^6 - 4x^3 + x^2 + x + 1 + x^3}$.



1

-0.0

Correct



2

-0.5

Some meaningful progress is made toward a correct answer, but there are still significant mistakes. The correct answer is 2.



3

-1.0

Little to no progress is made toward a correct answer.
The correct answer is 2.

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Q1.2 (b)

1 point

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(b) Evaluate $\lim_{x \rightarrow 1} \frac{|x^2 - 1|}{x^2 - 3x + 2}$.



1

-0.0

Correct



2

-0.5

Some meaningful progress is made toward a correct answer, but there are still significant mistakes. The correct answer is 2.



3

-1.0

Little to no progress is made toward a correct answer.
The correct answer is 2.

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Q1.3 (c)

2 points

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(c) Show that there are at least two x -values where the graphs of $f(x) = 3^{1-x^2} - 1$ and $g(x) = 2 - \cos(x)$ intersect. Be sure to fully justify your reasoning.



1

+2.0

Correct



2

+0.5

Some justification is provided that $f(x)$ and $g(x)$ are continuous.



3

+0.5

Some $a < b$ are found such that $f(a) - g(a) < 0$ and $f(b) - g(b) > 0$.



4

+0.5

Some $c < d$ are found such that $f(c) - g(c) > 0$ and $f(d) - g(d) < 0$.



5

+0.5

The Intermediate Value Theorem is cited as the reason for there existing intersections of $f(x)$ and $g(x)$ within the found intervals.



6

+0.0

Little to no progress is made toward a correct answer.

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Q2 A2

4 points

Q2.1 (a)

1.5 points

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(a) Use the limit definition of the derivative to find the derivative of $f(x) = \frac{-3}{x^2}$. You must use the limit definition of the derivative to get any credit for this problem.



1

+1.5

Correct. The answer is $\frac{6}{x^3}$.



2

+0.5

The correct limit is setup.



3

+0.5

Some meaningful progress is made toward evaluating the limit, but there are still significant mistakes.

⋮ 4 +1.0

The limit is correctly evaluated.

⋮ 5 +0.0

Little to no progress is made toward a correct answer.

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Q2.2 (b)

2.5 points

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(b) Consider the piecewise function

$$h(x) = \begin{cases} x + 1 & x \leq 1 \\ 2\sqrt{x} & x > 1 \end{cases}$$

Use the limit definition of the derivative to determine if $h(x)$ is differentiable at $x = 1$. You must use the limit definition of the derivative to get any credit for this problem.

⋮ 1 +2.5

Correct

⋮ 2 +0.5

Correct left-hand limit is setup: $\lim_{h \rightarrow 1^-} \frac{h(x) - h(1)}{x - 1}$.

**3****+0.5**

Left-hand limit is evaluated to 1.

**4****+0.5**

Correct right-hand limit is setup: $\lim_{h \rightarrow 1^+} \frac{h(x) - h(1)}{x - 1}$.

**5****+0.5**

Right-hand limit is evaluated to 1.

**6****+0.5**

Correctly concludes that $h'(1)$ does exist.

**7****+0.0**

No credit

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Q3 A3

4 points

Q3.1 (a)

2 points

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(a) Find the derivative of $f(x) = x^{\cos(\ln(x))}$. You do not need to simplify your answer.

⋮ 1 **+2.0**

Correct. The answer is

$$x^{\cos(\ln(x))} \left[-\sin(\ln(x)) \cdot \frac{1}{x} \ln(x) + \cos(\ln(x)) \cdot \frac{1}{x} \right]$$

⋮ 2 **+0.5**

The natural logarithm is used in some correct capacity to transform the function into a product.

⋮ 3 **+0.5**

Some meaningful progress is made toward evaluating the derivative of the product $\cos(\ln(x)) \cdot \ln(x)$ but there are still significant mistakes.

⋮ 4 **+1.0**

The product rule is used correctly to find the derivative of $\cos(\ln(x)) \cdot \ln(x)$.



5

+0.5

The chain rule is used (if the original function was transformed using $e^{\ln(f(x))}$), or both sides of the equation

$$\frac{f'(x)}{f(x)} = \left[-\sin(\ln(x)) \cdot \frac{1}{x} \ln(x) + \cos(\ln(x)) \cdot \frac{1}{x} \right]$$

are multiplied by $f(x)$ to solve for $f'(x)$.



6

+0.0

No meaningful progress is made.

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Q3.2 (b)

2 points

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(b) Find the derivative of $g(x) = \sqrt[3]{\frac{e^x + a}{2x^3 + ax}}$ where a is a constant. You do not need to simplify your answer.



1

+2.0

Correct



2

+0.5

Some meaningful progress is made toward using the chain rule appropriately, but there are still significant

mistakes.

⋮ 3 +1.0

The chain rule is successfully used.

⋮ 4 +0.5

Some meaningful progress is made toward evaluating the derivative of the quotient $\frac{e^x + a}{2x^3 + ax}$, but there are still significant mistakes.

⋮ 5 +1.0

The derivative of the quotient $\frac{e^x + a}{2x^3 + ax}$ is successfully computed.

⋮ 6 +0.0

Little to no progress is made.

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Q4 A4

4 points

Q4.1 (a)

2 points

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(a) Find the slope of the implicit curve $\tan(xy) = \frac{y}{1+x^2} - \frac{\pi}{6} + \sqrt{3}$ at $\left(1, \frac{\pi}{3}\right)$. You do not need to simplify your answer.

⋮ **+2.0**
Correct

⋮ **+0.5**

Some meaningful progress is made toward implicit differentiation of the equation, but there are still significant mistakes. The correct answer is

$$\sec^2(xy) \left(y + x \frac{dy}{dx} \right) = \frac{(1+x^2) \frac{dy}{dx} - 2xy}{(1+x^2)^2}.$$

⋮ **+1.0**

The implicit differentiation is done correctly. The correct

answer is $\sec^2(xy) \left(y + x \frac{dy}{dx} \right) = \frac{(1+x^2) \frac{dy}{dx} - 2xy}{(1+x^2)^2}.$

⋮ **+0.5**

Some meaningful progress is made toward solving the equation for $\frac{dy}{dx} = -\frac{3\pi}{7}$, but there are still significant mistakes.

5 +1.0

The equation is correctly solved for $\frac{dy}{dx}$.

6 +0.0

Little to no progress is made toward a solution.

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Q4.2 (b)

2 points

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(b) Use linearization to approximate $\frac{1}{\sqrt[3]{7.9}}$. You do not need to simplify your answer.

1 +2.0

Correct. The answer is $\frac{241}{480}$.

2 +0.5

$$f(8) = \frac{1}{2} \text{ is found.}$$

⋮ 3 **+0.5**

$$f'(8) = -\frac{1}{48} \text{ is found.}$$

⋮ 4 **+0.5**

$$L(x) = -\frac{1}{48}(x - 8) + \frac{1}{2} \text{ is found.}$$

⋮ 5 **+0.5**

$$L(7.9) = \frac{241}{480} \text{ is found.}$$

⋮ 6 **+0.0**

Little to no progress is made toward a solution.

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Q5 A5

4 points

Q5.1 (a)

1.5 points

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(a) A sphere is shrinking so that its volume is decreasing at a rate of $16 \text{ cm}^3/\text{min}$. How fast is the sphere's radius decreasing when the volume is $\frac{32\pi}{3} \text{ cm}^3$? Include units in your answer. *Hint:* The volume of a sphere of radius r is $V = \frac{4}{3}\pi r^3$.



1

+1.5

Correct. The answer is $-\frac{1}{\pi} \text{ cm/min}$.



2

+0.5

The correct units are reported.



3

+0.5

The time derivative of $V = \frac{4}{3}\pi r^3$ is taken correctly.



4

+0.5

$r = 2$ is found correctly and used to compute $\frac{dr}{dt} = -\frac{1}{\pi} \text{ cm/min}$.



5

+0.0

Little to no progress is made toward a solution.

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Q5.2 (b)

2.5 points

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(b) A person is flying a kite with a constant length of string, and the wind begins to cause the kite's height to decrease at a constant rate of 2 feet per second. At what rate is the kite's angle of elevation changing when the horizontal distance from the person to the point on the ground directly below the kite is 100 feet. Include units in your answer.

⋮ 1 +2.5

Correct. The answer is -0.02 radians per second.

⋮ 2 +0.5

The correct units are reported.

⋮ 3 +0.5

The correct equation is setup relating the variables of interest. One possible equation is $\sin \theta = \frac{y}{L}$.

⋮ 4 +0.5

Some meaningful progress is made toward taking the correct time derivative of the equation, but there are still significant mistakes.

⋮ 5 +1.0

The time derivative of the equation is taken correctly.



6

+0.5

The correct values for the given variables are found and plugged into the time derivative equation to solve for $\frac{d\theta}{dt}$.



7

+0.0

Little to no progress is made toward a solution.

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